

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 07: Tuesday 01 September 2026

(Week 4)

Remark. Q06.4 to Q06.6 are leftover exercises from the Friday class in Week 3.

Remark. Q07.1 to Q07.9 are on Section 13.1-13.3.

Questions on Section 13.7 will be set for this Friday.



Q06.3.5. (Supplementary Exercise).

Consider the function $f(x, y) = x^2 - y$. Sketch the level curves $f(x, y) = k$ for $k = -2, -1, 0, 1, 2$.

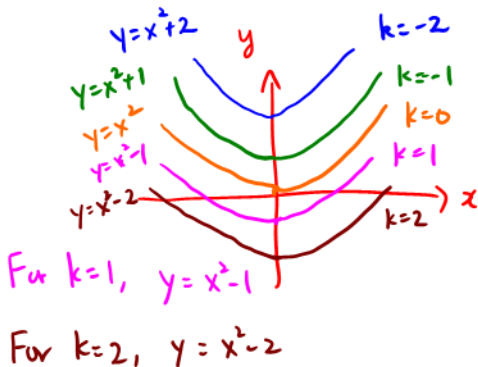
For a constant k , the k -level curve of f is given by $f(x, y) = k$

$$x^2 - y = k$$

For $k = -2$, $x^2 - y = -2$
 $y = x^2 + 2$

For $k = -1$, $x^2 - y = -1$
 $y = x^2 + 1$

For $k = 0$, $x^2 - y = 0$
 $y = x^2$



For $k = 1$, $y = x^2 - 1$

For $k = 2$, $y = x^2 - 2$

Function of 3 variables $f(x, y, z)$

The graph of $f(x, y, z)$ can be written as

$$w = f(x, y, z). \quad (*)$$

For example, for the function $f(x, y, z) = x^2 + y^2 + z^2$, its graph is:

$$w = x^2 + y^2 + z^2. \quad (*)$$

Unlike functions of two variables, it is very difficult to visualize a function $f(x, y, z)$ of three variables by its graph.

That graph $(*)$ is a three-dimensional object ('solid') lying in a four-dimensional space $\mathbb{R}^4$!!!

Again we do gain some insight into $f(\mathbf{x}, \mathbf{y}, \mathbf{z})$ by examining its level surfaces (counterparts of level curves in two-variable case).

A **level surface** of $f(\mathbf{x}, \mathbf{y}, \mathbf{z})$ is the surface in $\mathbb{R}^3$ defined by the equation $f(\mathbf{x}, \mathbf{y}, \mathbf{z}) = k$ for some constant k .

By studying the level surfaces corresponding to several different values of k , we may get some understanding of the function $f(\mathbf{x}, \mathbf{y}, \mathbf{z})$.

$$k=0$$

$$x^2 + y^2 + z^2 = 0$$

$$\Leftrightarrow (x, y, z) = (0, 0, 0)$$

Example. Consider the function

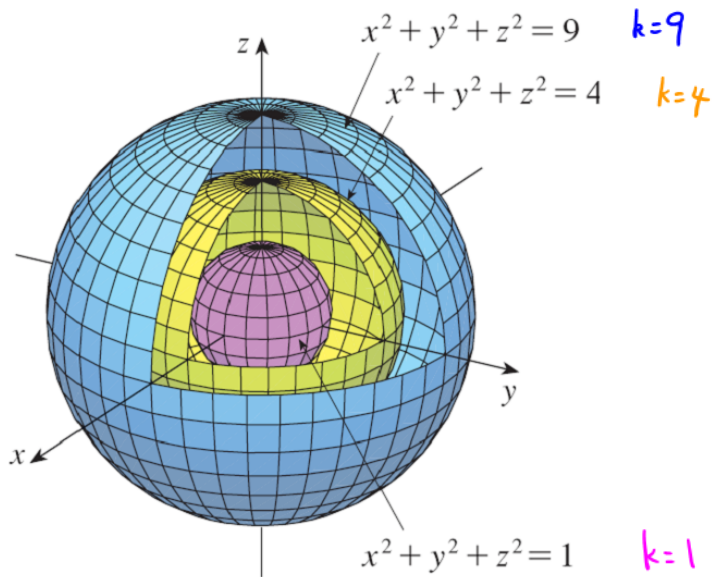
$$f(x, y, z) = x^2 + y^2 + z^2.$$

The level surfaces are:

$$x^2 + y^2 + z^2 = k, \quad k \geq 0$$

where $k \geq 0$.

For each $k \geq 0$, the level surface is a sphere in $\mathbb{R}^3$ with radius $\sqrt{k}$ centered at the origin $(0, 0, 0)$.



Q06.4. Describe the level sets of the following function geometrically:

$$f(x, y, z) = \sqrt{16 - x^2 - y^2 - z^2}. \quad = k$$

Solution: Note that for any $(x, y, z) \in \mathbb{R}^3$, since $x^2 + y^2 + z^2 \geq 0$, we have

$$0 \leq f(x, y, z) \leq 4.$$

Hence for the equation $f(x, y, z) = k$ to have solutions, we must have

$$\underline{0 \leq k \leq 4.}$$

Now under the condition that $0 \leq k \leq 4$, we have

$$f(x, y, z) = k$$

$$\iff \sqrt{16 - x^2 - y^2 - z^2} = k$$

$$\iff 16 - x^2 - y^2 - z^2 = k^2$$

$$\iff x^2 + y^2 + z^2 = 16 - k^2$$

$$\iff \underline{x^2 + y^2 + z^2 = (\sqrt{16 - k^2})^2}.$$

Hence for all $0 \leq k \leq 4$, the k -level set of f is a sphere of radius $\sqrt{16 - k^2}$ and centered at the origin $(0, 0, 0)$.

Q06.5. Evaluate the following limits:

(i) $\lim_{(x,y) \rightarrow (0,\pi/4)} \sec x \tan(x + y).$

(ii) $\lim_{(x,y) \rightarrow (2,0)} \frac{\sqrt{2x - y} - 2}{2x - y - 4}.$

[Remark: For (ii), you may find the identity

$a^2 - b^2 = (a + b)(a - b)$ useful.]

Q06.5.

Solution: (i)

// 1 · //

$$\lim_{(x,y) \rightarrow (0, \pi/4)} \sec x \tan(x+y) = \sec 0 \tan(0 + \frac{\pi}{4}) = 1 \cdot 1 = \underline{1}.$$

$\neq$
 $(0, \frac{\pi}{4})$

Remark: Roughly, (i) means that for those points where (x, y) is very close to $(0, \frac{\pi}{4})$ (but not exactly $(0, \frac{\pi}{4})$) and $\sec x \tan(x+y)$ is defined, $\sec x \tan(x+y)$ will be very close to 1. (See Thomas' Calculus, Section 13.2.)

$$(ii) \lim_{(x,y) \rightarrow (2,0)} \frac{\sqrt{2x-y}-2}{2x-y-4}$$

$$\frac{\sqrt{2 \cdot 2 - 0} - 2}{2 \cdot 2 - 0 - 4} = \frac{0}{0}$$

[Remark: For (ii), you may find the identity

indeterminate

$a^2 - b^2 = (a + b)(a - b)$ useful.]

Solution:

$$\lim_{(x,y) \rightarrow (2,0)} \frac{\sqrt{2x-y}-2}{2x-y-4}$$

$$\frac{a-b}{a^2-b^2} = \frac{a-b}{(a+b)(a-b)}$$

$$\begin{aligned} &= \lim_{(x,y) \rightarrow (2,0)} \frac{\cancel{\sqrt{2x-y}-2}}{(\sqrt{2x-y}+2)(\cancel{\sqrt{2x-y}-2})} \\ &\xrightarrow[(2x-y-4 \neq 0)]{(x,y) \neq (2,0)} = \lim_{(x,y) \rightarrow (2,0)} \frac{1}{\sqrt{2x-y}+2} \end{aligned}$$

$$= \frac{1}{\sqrt{2 \cdot 2 - 0} + 2} = \frac{1}{4}$$

Q06.6. Does the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{y}{x^2}$ exist? Justify your answer.

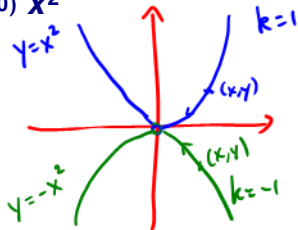
"0"
0
indeterminate

Solution:

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=x^2}} \frac{y}{x^2} = \lim_{(x,y) \rightarrow (0,0) \text{ along } y=x^2} 1 = 1.$$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=-x^2}} \frac{y}{x^2} = \lim_{(x,y) \rightarrow (0,0) \text{ along } y=-x^2} -1 = -1.$$

Thus $\lim_{(x,y) \rightarrow (0,0)} \frac{y}{x^2}$ does not exist.



$$\begin{aligned} k &= -1 \\ \frac{y}{x^2} &= -1 \\ y &= -x^2 \\ (x \neq 0, \\ (x,y) &\neq (0,0)) \end{aligned}$$

$$f(x,y) = \frac{y}{x^2}$$

Look at
level curves:
 $k=1$:

$$\begin{aligned} \frac{y}{x^2} &= 1 \\ y &= x^2 \\ (x \neq 0, \\ (x,y) &= (0,0)) \end{aligned}$$

Q07.1. Consider the function

$$f(x, y) = \begin{cases} \sqrt{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 1 & \text{if } (x, y) = (0, 0). \end{cases}$$

(i) Find $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$.

(ii) Is the function $f(x, y)$ continuous at $(0, 0)$? Justify your answer.

Solution: (i)

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \neq (0,0)}} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} = \sqrt{0^2 + 0^2} = \underline{\underline{0}}.$$

(ii)

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 \neq f(0,0) = 1.$$

Thus $f(x,y)$ is not continuous at $(0,0)$.

Continuity (see Thomas' Calculus, Section 13.2.)

Recall: A function $f(x, y)$ is **continuous** at a point (a, b) if $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$.

Remark. Roughly speaking, $f(x, y)$ is continuous at a point (a, b) means that the values of $f(x, y)$ at points (x, y) near (a, b) all become very close to $f(a, b)$ when (x, y) is very close to (a, b) , so that there is no sudden jump in the values of $f(x, y)$ at the point (a, b) .

Example. The function $f(x, y) = \sqrt{x^2 + y^2}$ is continuous for all (x, y) in the xy -plane.

Recall that for a function f of a single variable, we define the derivative function as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

for any values of x for which the limit exists.

At a particular point $x = a$, we interpret $f'(a)$ as the instantaneous rate of change of f with respect to x at that point. $(x=a)$.

We want to generalize the notion of derivative to functions of more than one variable.

The idea is to 'vary' one variable and keep other variable(s) fixed.

δ delta

$$\frac{df}{dx}$$

∂ rounded d

For a function $f(\mathbf{x}, \mathbf{y})$, its partial derivatives are the functions f_x and f_y defined by:

$$\frac{\partial f}{\partial x} = f_x(\mathbf{x}, \mathbf{y}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h, \mathbf{y}) - f(\mathbf{x}, \mathbf{y})}{h},$$

vary x ,
keep y
fixed

$$f_y(\mathbf{x}, \mathbf{y}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x}, \mathbf{y} + h) - f(\mathbf{x}, \mathbf{y})}{h}.$$

$$f(x+h, y) = (x+h)^2 y$$

Example. Let $f(x, y) = x^2 y$. Then

$$\begin{aligned} \frac{\partial f}{\partial x} &= f_x(x, y) = \lim_{h \rightarrow 0} \frac{(x+h)^2 y - x^2 y}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{x^2 y} + 2x\cancel{h}y + h^2 y - \cancel{x^2 y}}{\cancel{h}} \\ &\stackrel{h \neq 0}{=} \lim_{h \rightarrow 0} (2xy + hy) = \underline{2xy}. \\ f_y(x, y) &= \lim_{h \rightarrow 0} \frac{x^2(\cancel{y} + h) - \cancel{x^2 y}}{h} \stackrel{=}{=} \lim_{h \rightarrow 0} x^2 = x^2. \end{aligned}$$

Alternatively, f_x is also denoted by $\frac{\partial f}{\partial x}$.

To compute f_x , one may simply do the following:

Treat (temporarily) the other variable y in $f(x, y)$ as a constant, and differentiate $f(x, y)$ with respect to the variable x .

Similarly, to compute f_y (or $\frac{\partial f}{\partial y}$), one may simply do this:

Treat (temporarily) the other variable x in $f(x, y)$ as a constant, and differentiate $f(x, y)$ with respect to y .

Remark. f_x and f_y are themselves functions of two variables x, y .

Q07.2. (a) Find the first partial derivatives of each of the following functions:

(i) $f(x, y) = 4x^3 - 3x^2y^2 + 2x$.

(ii) $f(x, y) = \cos(3x^2y + 5y)$.

(iii) $f(x, y) = y \ln(4x + 3y)$.

(iv) $f(x, y, z) = y^2z + x^3yz^3$.

(b) (i) For the function in (a) (iii), find also the values of the first partial derivatives at the point **(2, 1)**.

(ii) For the function in (a) (iv), find also the values values of the first partial derivatives at the point **(1, 2, 4)**.

Solution: (i) $f(x, y) = 4x^3 - 3x^2y^2 + 2x$.

$$f_x = \frac{\partial f}{\partial x} = 12x^2 - 6y^2x + 2$$

$$\begin{aligned} f_y &= \frac{\partial f}{\partial y} = 0 - 6x^2y + 0, \\ &= -6x^2y. \end{aligned}$$

(ii) $f(x, y) = \cos(3x^2y + 5y)$.

$$\begin{aligned} f_x &= -\sin(3x^2y + 5y) \cdot (6xy + 0) \\ &= -6xy \sin(3x^2y + 5y). \end{aligned}$$

$$\begin{aligned} f_y &= -\sin(3x^2y + 5y) \cdot (3x^2 + 5) \\ &= -(3x^2 + 5) \sin(3x^2 + 5y). \end{aligned}$$

(iii) $f(x, y) = y \ln(4x + 3y)$.

(b) Find also $f_x(2, 1)$ and $f_y(2, 1)$.

$$f_x = y \cdot \frac{1}{4x+3y} \cdot (4+0) = \frac{4y}{4x+3y}$$

$$\begin{aligned} f_y &= 1 \cdot \ln(4x+3y) + y \cdot \frac{1}{4x+3y} \cdot (0+3) \\ &= \ln(4x+3y) + \frac{3y}{4x+3y}. \end{aligned}$$

$$f_x(2,1) = \frac{4 \cdot 1}{4 \cdot 2 + 3 \cdot 1} = \frac{4}{11}.$$

$$f_y(2,1) = \ln(4 \cdot 2 + 3 \cdot 1) + \frac{3 \cdot 1}{4 \cdot 2 + 3 \cdot 1} = \ln 11 + \frac{3}{11}.$$

(iv) $f(x, y, z) = y^2z + x^3yz^3$.

(b) Find also $f_x(1, 2, 4)$, $f_y(1, 2, 4)$ and $f_z(1, 2, 4)$.

Solution:

$$\frac{\partial f}{\partial x} = f_x = 0 + 3x^2yz^3 = 3x^2yz^3.$$

$$\frac{\partial f}{\partial y} = f_y = 2yz + x^3z^3.$$

$$\frac{\partial f}{\partial z} = f_z = y^2 + 3x^3yz^2.$$

(b)

$$f_x(1, 2, 4) = 3 \cdot 1^2 \cdot 2 \cdot 4^3 = 384.$$

$$f_y(1, 2, 4) = 2 \cdot 2 \cdot 4 + 1^3 \cdot 4^3 = 80.$$

$$f_z(1, 2, 4) = 2^2 + 3 \cdot 1^3 \cdot 2 \cdot 4^2 = 100.$$

Q07.3. Using implicit differentiation, find $\frac{\partial \mathbf{z}}{\partial \mathbf{x}}$ and $\frac{\partial \mathbf{z}}{\partial \mathbf{y}}$ if

$$\mathbf{x}^3 + 2\mathbf{y}^3 + 6\mathbf{x}\mathbf{z}^3 - 2 = 59\mathbf{z}.$$

[Remark: Here we think of $\mathbf{z}$ as a function of the two independent variables $\mathbf{x}$, $\mathbf{y}$ defined implicitly by the above equation. In other words, we think of an implicitly defined function $\mathbf{z}(\mathbf{x}, \mathbf{y})$.]

$$\mathbf{z} = \mathbf{z}(\mathbf{x}, \mathbf{y})$$

$$\frac{\partial}{\partial x}(z^3) = 3z^2 \cdot \frac{\partial z}{\partial x} \quad \Bigg| \quad z = z(x, y)$$

Solution: Given that

$$x^3 + \underline{2y^3} + \underline{6xz^3} - 2 = 59z. \quad (1)$$

Regarding z as a function of x , y , and taking partial derivatives of both sides of (1) with respect to x , we get

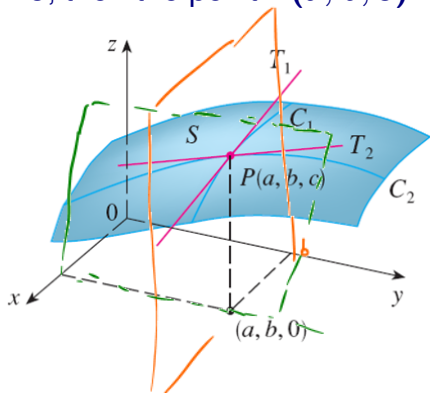
$$\begin{aligned} & \underline{3x^2} + \underline{0} + \underline{6 \cdot z^3} + \underline{6x \cdot 3z^2 \cdot \frac{\partial z}{\partial x}} + \underline{0} = \underline{59 \frac{\partial z}{\partial x}} \\ \Rightarrow & \underline{3x^2 + 6z^3} = (59 - 18xz^2) \frac{\partial z}{\partial x} \\ \Rightarrow & \frac{\partial z}{\partial x} = \frac{3x^2 + 6z^3}{59 - 18xz^2}. \end{aligned}$$

$$x^3 + 2y^3 + 6xz^3 - 2 = 59z. \quad (1)$$

Similarly, differentiating both sides of (1) with respect to y , we get

$$\begin{aligned} 0 + 6y^2 + 6x \cdot 3z^2 \cdot \frac{\partial z}{\partial y} + 0 &= 59 \frac{\partial z}{\partial y} \\ \implies 6y^2 &= (59 - 18xz^2) \frac{\partial z}{\partial y} \\ \implies \frac{\partial z}{\partial y} &= \frac{6y^2}{59 - 18xz^2}. \end{aligned}$$

Geometric interpretation of f_x and f_y : Recall that the equation $z = f(x, y)$ represents a surface $\mathbf{S}$ (the graph of f). If $f(\mathbf{a}, \mathbf{b}) = \mathbf{c}$, then the point $\mathbf{P}(\mathbf{a}, \mathbf{b}, \mathbf{c})$ lies on $\mathbf{S}$.



$$f_x(a, b)$$

$$f_y(a, b)$$

By fixing $y = b$, we are restricting our attention to the curve $\mathbf{C}_1$ in which the vertical plane $y = b$ intersects $\mathbf{S}$.

Also, the vertical plane $x = a$ intersects $\mathbf{S}$ in a curve $\mathbf{C}_2$.

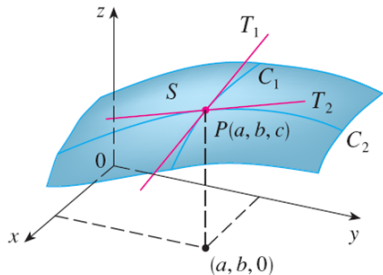
- C_1 is the graph of the function $g(x) = f(x, b)$.

So, the slope of its tangent T_1 at P is: $g'(a) = f_x(a, b)$.

- C_2 is the graph of the function $h(x) = f(a, y)$.

So, the slope of its tangent T_2 at P is: $h'(b) = f_y(a, b)$.

Summary: Geometrically, $f_x(a, b)$ is the slope of the tangent line at $P(a, b, c)$ to the curve C_1 , which is the intersection of S with the vertical plane $y = b$.



Similar geometrical interpretation holds for $f_y(a, b)$.

For a function $f(\mathbf{x}, \mathbf{y})$, its partial derivatives f_x and f_y are also functions of two variables $\mathbf{x}, \mathbf{y}$.

So, we can consider partial derivatives of f_x and f_y :

$$(f_x)_x, (f_x)_y, (f_y)_x, (f_y)_y.$$

These are called the second partial derivatives of f .

If $\mathbf{z} = \mathbf{f}(\mathbf{x}, \mathbf{y})$, we use the following notation: $(\partial_x)^2$

$$(f_x)_x = f_{xx} = \frac{\partial}{\partial \mathbf{x}} \left(\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right) = \frac{\partial^2 \mathbf{f}}{\partial \mathbf{x}^2} = \frac{\partial^2 \mathbf{z}}{\partial \mathbf{x}^2},$$

$$(f_x)_y = \underline{f_{xy}} = \frac{\partial}{\partial \mathbf{y}} \left(\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right) = \frac{\partial^2 \mathbf{f}}{\underline{\partial \mathbf{y} \partial \mathbf{x}}} = \frac{\partial^2 \mathbf{z}}{\partial \mathbf{y} \partial \mathbf{x}},$$

$$(f_y)_x = f_{yx} = \frac{\partial}{\partial \mathbf{x}} \left(\frac{\partial \mathbf{f}}{\partial \mathbf{y}} \right) = \frac{\partial^2 \mathbf{f}}{\partial \mathbf{x} \partial \mathbf{y}} = \frac{\partial^2 \mathbf{z}}{\partial \mathbf{x} \partial \mathbf{y}},$$

$$(f_y)_y = f_{yy} = \frac{\partial}{\partial \mathbf{y}} \left(\frac{\partial \mathbf{f}}{\partial \mathbf{y}} \right) = \frac{\partial^2 \mathbf{f}}{\partial \mathbf{y}^2} = \frac{\partial^2 \mathbf{z}}{\partial \mathbf{y}^2}.$$

Thus, the notation f_{xy} means that we first differentiate with respect to $\mathbf{x}$ and then with respect to $\mathbf{y}$.

In computing f_{yx} , the order is reversed.

Q07.4. Consider the function $f(x, y) = \sin(x + y^2)$.

(i) Evaluate f_x , f_y , f_{xx} , f_{xy} , f_{yx} , f_{yy} .

(ii) What do you observe about f_{xy} and f_{yx} ?

$$(ii) \quad f_{xy} = f_{yx}.$$

Solution:(i)

$$(ii) f_{xy} = f_{yx}.$$

$$f(x, y) = \sin(x + y^2)$$

$$f_x = \cos(x + y^2) \cdot 1 = \cos(x + y^2),$$

$$f_y = \cos(x + y^2) \cdot 2y = 2y \cos(x + y^2)$$

$$\begin{aligned} f_{xy} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (\cos(x + y^2)) \\ &= -\sin(x + y^2) \cdot 2y \\ &= -2y \sin(x + y^2). \end{aligned}$$

$$\begin{aligned} f_{yx} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (2y \cos(x + y^2)) \\ &= -2y \sin(x + y^2) \cdot 1 \end{aligned}$$

$$f_{xx} = -\sin(x + y^2), \quad f_{yy} = 2\cos(x + y^2) - 4y^2 \sin(x + y^2)$$

Clairaut's Theorem: Suppose $f(x, y)$ is defined on an open region $D \subseteq \mathbb{R}^2$ that contains (a, b) .

If the functions f_x , f_y , f_{xy} and f_{yx} are continuous on D , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

(See e.g. Thomas' Calculus, Section 13.3, p. 795.)

Q07.5. Verify the identity $\underline{f_{xy} = f_{yx}}$ for the function

$$f(x, y) = x^3 y^4 + e^{x+2y}.$$

Solution:

$$f_x = 3x^2 y^4 + e^{x+2y} \cdot 1 = 3x^2 y^4 + e^{x+2y},$$

$$\begin{aligned} f_{xy} &= (f_x)_y = \frac{\partial}{\partial y}(3x^2 y^4 + e^{x+2y}) \\ &= 12x^2 y^3 + e^{x+2y} \cdot 2 = 12x^2 y^3 + 2e^{x+2y}. \end{aligned}$$

$$f_y = x^3 \cdot 4y^3 + e^{x+2y} \cdot 2 = 4x^3 y^3 + 2e^{x+2y}.$$

$$\begin{aligned} f_{yx} &= (f_y)_x = \frac{\partial}{\partial x}(4x^3 y^3 + 2e^{x+2y}) \\ &= 12x^2 y^3 + 2e^{x+2y} \cdot 1 = 12x^2 y^3 + 2e^{x+2y}. \end{aligned}$$

Thus $f_{xy} = 12x^2 y^3 + 2e^{x+2y} = f_{yx}$.